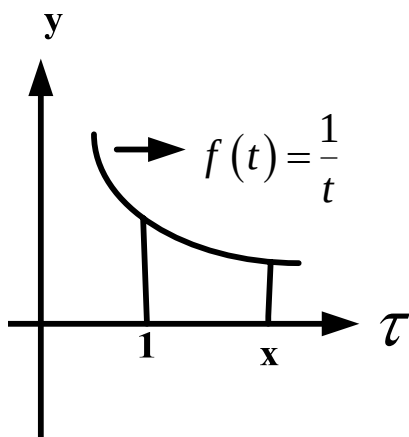


§5-6 The Logarithm Defined as an Integral

Homework : 3,4.

Definition :



定義： $\ln x = \int_1^x \frac{1}{t} dx, x > 0$

(I) Properties of $\ln x$.

(i) $\ln 1 = 0$, $\ln x > 0$, if $x > 1$; $\ln x < 0$ if $x < 1$.

(ii) $\frac{d}{dx} \ln x = \frac{1}{x}$.

(iii) $\ln xy = \ln x + \ln y$.

proof of (iii)

$$\text{Let } f(x) = \ln ax, a > 0 \Rightarrow f(x) = \int_1^{ax} \frac{1}{t} dt$$

$$\Rightarrow f'(x) = \frac{a}{ax} = \frac{1}{x} \Rightarrow \ln ax = \ln x + C$$

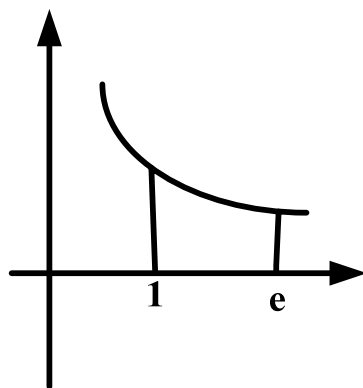
pick x=1

$$\Rightarrow \ln a = \ln 1 + C \Rightarrow C = \ln a$$

$$\Rightarrow \ln ax = \ln x + \ln a.$$

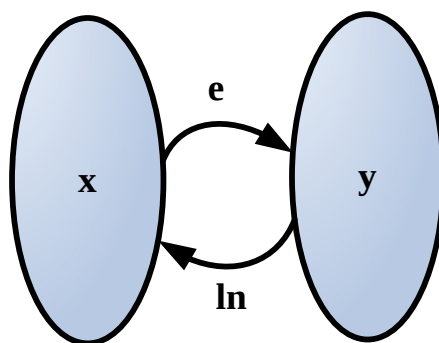
(II) Define e to be such that

$$\ln e = 1.$$



(III) Exponential Functions : $e^x = y \Leftrightarrow \ln y = x$

(Inverse function of $\ln$)



$$\Rightarrow \frac{1}{y} y' = 1 \Rightarrow y' = y \Rightarrow \frac{d}{dx} e^x = e^x.$$

(IV) (i) General Exponential Functions.

$$a^x = (e^{\ln a})^x = e^{x \ln a} \Rightarrow \frac{d}{dx} a^x = (\ln a) a^x.$$

(ii) General Logarithm (Inverse function of $f(x) = a^x$)

$$\begin{aligned} \log_a x = y &\Leftrightarrow a^y = x \Rightarrow (\ln a) a^y y' = 1 \\ \Rightarrow y' &= a^y \left(\frac{1}{\ln a} \right) \Rightarrow \frac{d}{dx} \log_a x = \left(\frac{1}{\ln a} \right) \frac{1}{x}. \end{aligned}$$

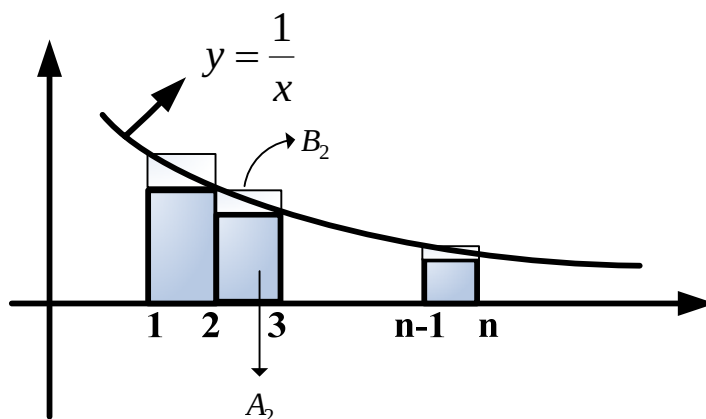
$$(V) \quad e = \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}}.$$

Remark : 有些微積分的書先講積分，再提微分。如此則可先定義 $y = \ln x$

再定義 $y = e^x$ 再定義 $y = a^x$ 再定義 $y = \log_a x$ 。這樣的好處可以避免需要解釋什麼是 2^π ?

Example 1 :

$$\frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} < \ln n < 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n-1}.$$



因為 $f(x) = \frac{1}{x}$ 為遞減函數，

$\frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} = \Sigma$ 小長方形 A_i 面積，且這些小長方形 A_i 皆在 $f(x) = \frac{1}{x}$ 底下。

$\frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n-1} = \Sigma$ 小長方形 B_i 面積，且這些小長方形 B_i 皆有部分

在 $f(x) = \frac{1}{x}$ 上面。